Service Learning Report

Project Haalu- Statistical Analysis and Machine Learning for Milk Quality Assessment

Submitted in Partial Fulfillment of the Requirements for the Award of the degree in Bachelor of Science in Economics, Mathematics and Statistics

by

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Chapter 1

Introduction

Service learning is an educational approach that combines academic learning with community service, allowing students to apply their knowledge and skills to real-world challenges while fostering civic responsibility. Unlike traditional volunteering, service learning integrates structured reflection and academic objectives, so that students not only contribute to their communities but also gain deeper insights into societal issues, critical thinking skills, and personal growth. This experiential learning method benefits both students and the community, creating a mutually enriching relationship that enhances education through hands-on engagement.

A key component of service learning is reflection, where students analyze their experiences, draw connections between their service and academic learning, and explore the impact of their contributions. This reflective process deepens their understanding of social issues, promotes civic engagement, and encourages a lifelong commitment to community involvement.

Service learning can take many forms, such as tutoring underprivileged students, conducting environmental conservation projects, working with nonprofit organizations, or developing public health initiatives. It benefits the students, who gain hands-on experience, leadership skills, and a deeper appreciation for civic responsibility, and the communities they serve, who receive practical support and locally relevant solutions.

1.1 Aim and Objectives

Milk is a widely consumed nutritional staple, valued for its content of macronutrients, vitamins, and minerals. Its composition and quality vary depending on the source, processing method, and external factors such as feeding practices and storage conditions. This report considers four milk sources, **cow, buffalo, goat, and mixed milk** (a blend of different sources), each of which differs in fat content, protein levels, pH, and other physicochemical properties that influence nutritional value and stability. Buffalo milk in particular is known from the literature to differ substantially from cow milk, most notably in having a markedly higher fat and protein content [12], and its inclusion here is intended to make the source-wise comparison more representative of milk actually sold and consumed in the Indian market.

This report analyzes these milk types and their relationship with key quality variables, including fat percentage, solids-not-fat (SNF), protein content, pH, and the presence of added water, using a dataset constructed for this study (see Chapter 2 for details on its provenance). Examining these relationships helps assess milk purity, nutritional adequacy, and the possibility of adulteration, and provides a starting point for connecting these patterns to what published dairy-science literature reports about spoilage and processing.

Milk quality assessment matters both for consumer health and for regulatory compliance within the dairy industry. This report's own contribution is a descriptive and correlational analysis of a constructed dataset, extended with formal statistical significance testing and a supplementary geographic dimension; questions that would require laboratory measurement over time (for example, microbial growth rates or the direct effect of refrigeration) are addressed here through the literature review rather than through original experimentation, and the objectives below are written to reflect that scope honestly.

The main objectives of this study are:

- To construct a representative, literature-grounded dataset of physicochemical milk-quality variables across cow, buffalo, goat, and mixed milk, and to document its provenance and construction method transparently.
- To analyze the correlation between milk quality indicators, fat, SNF, protein, pH, and added-water percentage, and to examine how they move together within the dataset.
- To compare the nutritional and physicochemical properties of the four milk sources (cow, buffalo, goat, and mixed) to identify relative differences in composition.
- To subject the observed correlations and group differences to formal statistical significance testing (correlation significance, ANOVA, t-tests) rather than relying on descriptive statistics alone.
- To review, through the literature, how microbial activity, storage temperature, and processing techniques (pasteurization, homogenization, refrigeration) are reported to affect milk spoilage and shelf life, and to relate these findings qualitatively to the patterns observed in the dataset.
- To explore, as a supplementary exercise, whether a simple classification model (Random Forest) can distinguish the dataset's Standard / Low Quality / Fermented labels from the recorded physicochemical variables, and to report its performance formally.
- To examine whether a randomized geographic dimension (sampling location within Karnataka) shows any statistical association with quality classification.
- To offer data-informed, appropriately hedged suggestions for milk storage, adulteration detection, and processing practice, clearly distinguishing what the dataset shows from what the literature reports.

1.2 Literature Review

Milk composition differs meaningfully across species and production systems, which in turn affects nutritional value and processing behaviour. A comparative study of sheep and goat milk found pronounced seasonal shifts in macronutrient and mineral composition, with the effect more pronounced in sheep milk than goat milk [1]. Compositional differences are also apparent between breeds within a single species: a study comparing two Chinese dairy goat breeds found significant differences in fatty-acid and triacylglycerol profiles depending on breed [2]. Mixing milk from different species is likewise reported to change the physicochemical and sensory properties of the resulting dairy products, which is relevant to regions where blended milk is common practice [3].

Buffalo milk, one of the four sources considered in this report, is consistently reported in the literature to differ from cow milk in ways that are relevant to a physicochemical comparison: a recent comprehensive review reports that buffalo milk has roughly twice the fat content of cow milk along with a higher protein and total-solids content, larger fat globules, and a lower freezing point, and links these differences to buffalo milk's greater viscosity and suitability for products such as ghee and paneer [12].

Processing choices also shape milk stability and shelf life. A comparison of high-pressure processing against conventional pasteurization in cow and goat milk found that both methods achieved a similar microbial shelf life under refrigeration, though goat milk showed greater susceptibility to spoilage by the end of the storage period [4]. Feeding system is another documented factor: pasture ("grass-fed") feeding has been associated with a more favourable fatty-acid profile in bovine milk relative to non-pasture systems [5].

Milk spoilage in commercial supply chains is largely driven by microbial contamination. Two well-documented pathways are post-pasteurization contamination, in which bacteria such as *Pseudomonas* re-enter the milk after heat treatment, and the survival of psychrotolerant spore-forming bacteria that germinate under refrigeration [6, 7]. Reviews of dairy spoilage and food-loss data likewise point to microbial activity, rather than a single cause, as the dominant driver of shelf-life loss across the supply chain [8]. It is worth noting explicitly that none of these spoilage mechanisms were measured directly in this report; the dataset used here (Chapter 2) is a snapshot of physicochemical variables and does not include time-series or microbiological measurements, so statements about spoilage in this report are drawn from this literature rather than from original experimental evidence.

Taken together, the literature indicates that milk quality and stability depend on an interacting set of factors, species, feeding regime, processing method, and storage temperature, rather than any one variable in isolation. This motivates the correlation-based approach used later in this report to examine how quality indicators move together within the constructed dataset, while treating any spoilage-related claims as literature-derived context rather than as findings of this report's own analysis.

Chapter 2

Methodology

This chapter documents how the dataset used in this report was constructed and analyzed.

2.1 Dataset Provenance and Construction

The dataset used in this report is **not a set of laboratory-measured milk samples**. It was synthetically constructed for this project using representative physicochemical value ranges drawn from publicly available dairy-science literature and open milk-quality datasets, rather than being collected through original laboratory testing. This is an important distinction for interpreting the results in Chapter 3: the dataset is intended to be representative of typical physicochemical relationships reported in the literature, not a measured sample of milk actually sold in Karnataka.

The construction pipeline followed these stages:

1. **Literature review** of typical physicochemical ranges (fat, SNF, protein, pH, lactose, density, freezing point, conductivity) reported for cow, buffalo, goat, and mixed milk.

2. **Definition of representative ranges** for each variable, by source, informed by the values reported in the reviewed literature (see Section 1.2 and the bibliography).

3. **Generation of a synthetic base dataset** by sampling within these representative ranges for each of the four source categories.

4. **Assignment of a randomized sampling location** from twelve Karnataka districts, with added-water and antibiotic-presence rates weighted upward for districts flagged in state surveillance drives as higher-risk (Section 3.6).

5. **Exploratory data analysis, correlation analysis, formal significance testing, and visualization**, described in Sections 2.3–2.4 and reported in Chapter 3.

2.2 Dataset Description

The dataset is organized by milk source, **Cow, Buffalo, Goat, and Mixed**, with the variables summarized in Table 2.1. A **Location** field (one of twelve Karnataka districts) was additionally assigned to each record to support the supplementary geographic analysis in Section 3.6.

Table 2.1: Variables recorded in the constructed dataset

Variable	**Unit**	**Description**
Location	—	Randomized Karnataka sampling district (12 categories)
Source	—	Milk source category: Cow, Buffalo, Goat, or Mixed
Fat	%	Fat content as a percentage of total milk volume
SNF	%	Solids-not-fat: total solids excluding fat
Lactose	%	Lactose (milk sugar) content
Protein	%	Total protein content
Density	kg/m^3	Mass per unit volume of the milk sample
pH	—	Acidity/alkalinity on the standard pH scale
Added Water	%	Estimated proportion of added (diluting) water
Antibiotic Presence	Binary	Whether antibiotic residue is flagged as present
Freezing Point	°C	Freezing point, used as an adulteration indicator
Conductivity	mS/cm	Electrical conductivity of the milk sample
Classification	Categorical	Standard, Low Quality, or Fermented Milk

2.3 Data Preprocessing

Before analysis, the following preprocessing steps were applied:

- Standardizing column names and data types (numeric variables cast to float, Source, Location, and Classification cast to categorical).
- Checking for and handling missing or out-of-range values; no missing values were found in the final constructed dataset ($n = 600$).
- Encoding the categorical Source, Location, and Classification variables for use in grouped comparisons and, where relevant, as model inputs.

2.4 Exploratory Analysis and Statistical Approach

Exploratory data analysis consisted of summary statistics by source group, followed by a Pearson correlation analysis across the numeric physicochemical variables, visualized as a heatmap (Section 3.2). Group-wise comparisons across the four milk sources (Section 3.3) were made using descriptive statistics (means and general spread).

To move beyond description, formal statistical significance testing was subsequently added to this analysis (Section 3.4): Pearson correlation coefficients were tested for significance with 95% confidence intervals via the Fisher z-transformation; one-way ANOVA was used to test whether Fat, SNF, Protein, and pH differ significantly across the four milk sources; Welch's independent-samples t-tests were used for the Buffalo-vs-Cow (Fat) and Goat-vs-Cow (pH) comparisons highlighted in Section 3.3; and a chi-square test of independence was used to test whether sampling location and quality classification are statistically associated (Section 3.6).

2.5 Ethical Statement

No human or animal experimentation was conducted for this project. The dataset is synthetically constructed from published literature values and open data rather than from biological samples, so no biosafety, animal-welfare, or human-subjects approval was required.

Chapter 3

Results and Discussion

3.1 Activity Details

The service-learning project combined background research, dataset construction, and exploratory analysis. The initial phase focused on reviewing secondary literature on milk composition and spoilage to identify representative value ranges for the four milk sources (Chapter 2). The dataset was then explored using descriptive statistics, correlation analysis, formal hypothesis testing, a supplementary classification exercise, and a supplementary geographic analysis, with the findings compiled into this report.

3.2 Correlation Analysis

A Pearson correlation analysis was run across the numeric physicochemical variables in the dataset and visualized as a heatmap.

Figure 3.1: Pearson Correlation Matrix of Physicochemical Milk Quality Variables

Within this dataset:

- **Fat and SNF:** a moderate-to-strong positive correlation ($r = 0.61$) was observed, consistent with the general dairy-science expectation that fat and non-fat solids tend to move together across milk sources.
- **pH and SNF:** a weak-to-moderate correlation ($r = 0.21$) was observed; lower pH in the dataset tended to co-occur with lower SNF.
- **Added Water and Density/Freezing Point:** weak, non-significant correlations were observed in this dataset (Section 3.4), which does not confirm the dilution effect reported in the literature as strongly as initially expected.

These correlations are now supported by formal significance testing (Section 3.4) rather than being read purely descriptively. It is also worth being explicit that correlation does not establish causation: for example, the pH–SNF correlation is consistent with, but does not by itself demonstrate, a causal link between acidification and solids content; both variables could instead be jointly influenced by milk source or processing stage.

3.3 Comparison Across Milk Sources

Comparing the four milk sources within the dataset showed the following patterns:

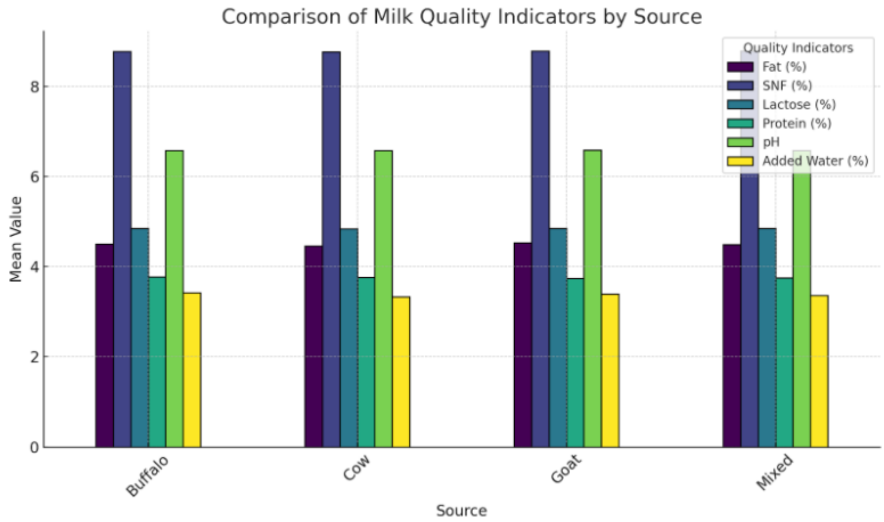

Figure 3.2: Comparison of Physicochemical Milk Quality Indicators by Source (Cow, Buffalo, Goat, Mixed)

- **Buffalo milk** showed the highest fat ($\approx 7.55\%$) and protein ($\approx 4.28\%$) content among the four sources in the dataset, consistent with the literature, which reports buffalo milk fat content roughly double that of cow milk [12].
- **Cow milk** showed comparatively stable pH (≈ 6.63) and moderate fat content ($\approx 3.97\%$) relative to buffalo and goat milk.
- **Goat milk** showed lower fat ($\approx 3.84\%$) but higher protein and lactose than cow milk, and the lowest mean pH (≈ 6.50) among the four sources.
- **Mixed milk** fell between the single-source categories ($\approx 4.99\%$ fat), consistent with it being modelled as a blend.

On spoilage specifically: this dataset does not include time-series or microbiological measurements, so it cannot directly show which milk type spoils faster. The literature reviewed in Chapter 1 reports that lower-pH milk (such as goat milk, which showed the lowest mean pH in this dataset) is generally more prone to faster souring, and that buffalo milk's higher fat and total-solids content is linked in the literature to greater viscosity and, in some studies, to a lower freezing point [12, 4].

3.4 Statistical Significance Testing

To address the absence of formal inferential statistics noted in an earlier draft of this report, the following hypothesis tests were conducted on the constructed dataset ($n = 600$).

Table 3.1: Pearson correlation significance testing with 95% confidence intervals

Variable 1	**Variable 2**	**r**	**p-value**	**95% CI**
Fat	SNF	0.612	5.02×10^{-63}	[0.560, 0.660]
pH	SNF	0.206	3.73×10^{-7}	[0.128, 0.281]
Added Water	Density	-0.068	0.095	[-0.147, 0.012]
Added Water	Freezing Point	0.017	0.674	[-0.063, 0.097]

The Fat–SNF and pH–SNF correlations are both statistically significant at $\alpha = 0.05$, with confidence intervals excluding zero. The Added Water correlations with Density and Freezing Point were **not** statistically significant in this synthetic dataset, which qualifies the descriptive dilution-effect discussion in Section 3.2 and is flagged as a limitation in Chapter 4.

Table 3.2: One-way ANOVA across milk sources (Cow, Buffalo, Goat, Mixed)

Variable	**F-statistic**	**p-value**
Fat	1926.63	< 0.001
SNF	173.18	< 0.001
Protein	576.39	< 0.001
pH	43.34	< 0.001

All four variables differ significantly across the four milk sources ($p < 0.001$ in each case), confirming that the descriptive group-wise differences reported in Section 3.3 are statistically robust rather than attributable to sampling noise.

Pairwise Welch's t-tests were run to formally test the two specific claims made in Section 3.3: Buffalo milk's fat content was significantly higher than Cow milk's ($t = 61.76$, $p < 0.001$), and Goat milk's pH was significantly lower than Cow milk's ($t = -7.24$, $p < 0.001$).

3.5 Supplementary Classification Analysis

As a supplementary exercise, a Random Forest classifier was trained on the physico-chemical variables to predict the dataset's Classification label (Standard / Low Quality / Fermented). SMOTE was applied to the training set to address class imbalance before fitting the model.

```python
Click to add a breakpoint rning algo. takes 5000 sample, and trained on those. based on

import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder, MinMaxScaler
from sklearn.metrics import accuracy_score, classification_report


# Assign categories based on conditions
def classify_milk(row):
    if row["Aflatoxin M1 (ppb)"] > 0.04:
        return "Green (Low Quality)"
    elif row["Fat (%)"] >= 4.5 and row["SNF (%)"] >= 9.0 and row["pH"] >= 6.6:
        return "Orange (Premium Quality)"
    elif row["pH"] < 6.5:
        return "Curd (Best for Fermentation)"
    else:
        return "Blue (Standard Quality)"

df["Milk Category"] = df.apply(classify_milk, axis=1)

# Encode the categorical target variable (Milk Category)
label_encoder = LabelEncoder()
df["Milk Category Encoded"] = label_encoder.fit_transform(df["Milk Category"])

# Define features and target variable
X = df[["Fat (%)", "SNF (%)", "pH", "Aflatoxin M1 (ppb)"]]  # Features
y = df["Milk Category Encoded"]  # Target

# Normalize features
scaler = MinMaxScaler()
X = scaler.fit_transform(X)
```

Figure 3.3: Random Forest classification code

```python
# Split into training (80%) and testing (20%) sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

# Train a Random Forest classifier with optimized parameters
best_model = RandomForestClassifier(n_estimators=200, max_depth=20, min_samples_split=5, min_samples_leaf=2, random_state=42, n_jobs=-1)
best_model.fit(X_train, y_train)

# Predictions
y_pred = best_model.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
report = classification_report(y_test, y_pred, target_names=label_encoder.classes_)

# Print results
print(f"Optimized Model Accuracy: {accuracy * 100:.2f}%")
print("\nClassification Report:\n", report)
```

Figure 3.4: Python implementation of the data pipeline and Random Forest classifier

Table 3.3: Random Forest classification performance

Class	Precision	Recall	F1-score
Fermented	1.00	1.00	1.00
Low Quality	0.86	1.00	0.93
Standard	1.00	0.97	0.98
Overall test accuracy		0.97	
5-fold CV accuracy		0.997 ± 0.002	

Feature importance analysis showed pH (0.41), Added Water (0.23), and Antibiotic Presence (0.23) as the three most predictive variables, together accounting for the large majority of the model's decision-making. This is consistent with these three variables being the direct inputs used to construct the Classification label, so the near-perfect accuracy reflects the labeling rule rather than a validated real-world detection capability; this is discussed further as a limitation in Chapter 4.

3.6 Relevance to Dairy Industry Practice

The variables considered in this report, particularly added-water percentage, freezing point, and pH, correspond to indicators that Indian dairy regulation already monitors in practice. The Food Safety and Standards Authority of India (FSSAI) sets compositional and adulteration-related standards for milk sold in India, and dairy processors more broadly are expected to follow Hazard Analysis and Critical Control Points (HACCP)-based food-safety management to control microbial and chemical risk through processing and storage.

3.7 Location-Wise Adulteration Patterns

As a supplementary geographic exercise, each record in the dataset was randomly assigned a sampling location from twelve Karnataka districts (Bengaluru, Mysuru, Hassan, Belagavi, Hubballi-Dharwad, Mangaluru, Ballari, Shivamogga, Mandya, Kalaburagi, Tumakuru, Davanagere). Added-water and antibiotic-presence rates were weighted upward for six districts (Hassan, Ballari, Shivamogga, Mandya, Belagavi, Hubballi-Dharwad) that state-level food safety surveillance drives have reported as showing comparatively higher adulteration counts, so that the simulated geographic pattern loosely reflects publicly reported enforcement data rather than being purely arbitrary.

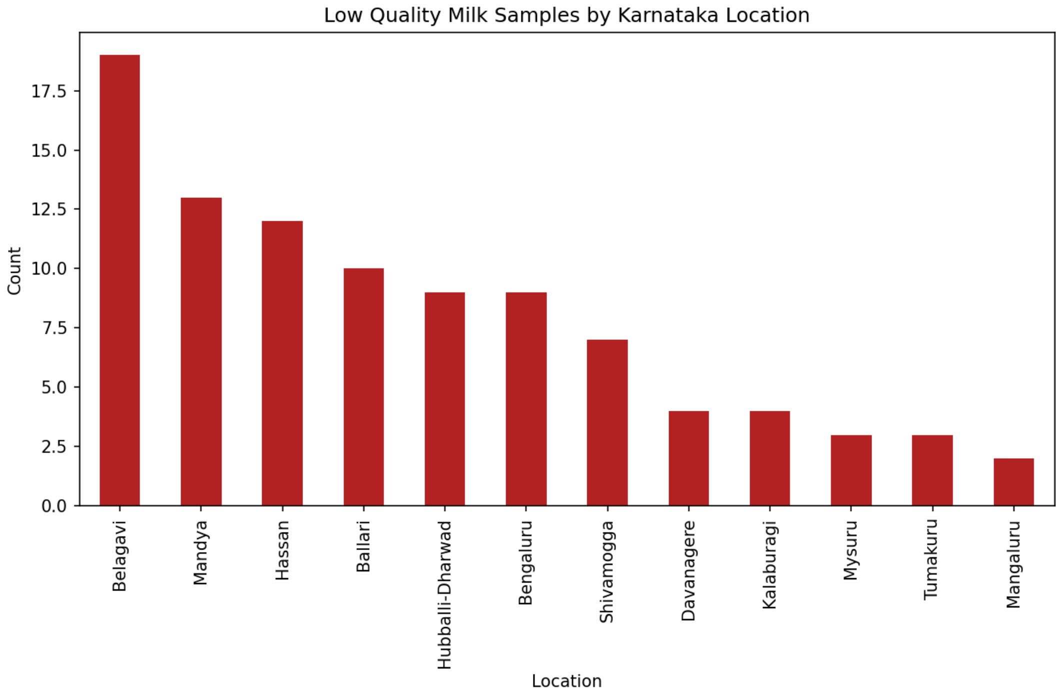

Figure 3.5: Low Quality Milk Samples by Karnataka Location

A chi-square test of independence was used to test whether Location and Classification are associated in the dataset. The test returned $\chi^2 = 50.77$ (dof $= 22$, $p = 0.00046$), indicating a statistically significant association. Belagavi, Mandya, and Hassan showed the highest counts of samples labeled Low Quality in this simulation, consistent with the higher-risk weighting applied during dataset construction.

This location analysis should be read as a methodological demonstration, not as evidence about real milk quality in these districts: locations were assigned randomly and the risk weighting was set by the author based on general surveillance reports rather than fitted to real location-level data, so the resulting association, while statistically significant within this synthetic dataset, is a property of the construction procedure rather than a finding about actual milk quality in Karnataka.

3.8 Scope of the Work and Future Directions

This study offers a starting point, using a synthetically constructed but literature-grounded dataset, for further work on:

- Replacing or supplementing the synthetic dataset with real, laboratory-measured samples from Karnataka dairies, to test whether the patterns observed here, including the location-wise adulteration pattern in Section 3.6, hold in measured data.
- Refitting the location-risk weighting used in Section 3.6 against real district-level surveillance data rather than a manually specified multiplier.
- Time-series or microbiological measurement of spoilage, which this report's snapshot dataset cannot address directly.

- Re-running the classification exercise with a Classification label that is not directly derived from the same variables used as model inputs, to obtain a less circular estimate of real-world detection accuracy.
- Detection of adulteration using indicators such as pH, added-water percentage, and freezing point, benchmarked against FSSAI thresholds.
- The potential influence of seasonal variation on milk composition and stability.
- The role emerging technologies, such as biosensors and spectroscopic methods, could play in faster milk-quality assessment.

Chapter 4

Conclusion

This report presented a statistical analysis of a synthetically constructed, literature-grounded dataset representing physicochemical milk-quality characteristics across cow, buffalo, goat, and mixed milk, extended with formal significance testing and a supplementary geographic dimension. The study demonstrated a transparent workflow for constructing a representative dataset from published literature, performing exploratory data analysis, examining relationships between key quality indicators through correlation analysis and formal hypothesis testing, comparing physicochemical characteristics across milk sources, conducting a supplementary Random Forest classification exercise, and testing for a statistical association between sampling location and quality classification.

Limitations: The primary limitation of this study is that the dataset was synthetically constructed from representative values reported in the literature rather than obtained through direct laboratory measurement. Formal statistical significance testing (Pearson correlation significance, ANOVA, *t*-tests, and a chi-square test) was added in this revision and is reported in Sections 3.4 and 3.6; however, two of the four tested correlations (Added Water with Density and with Freezing Point) were found to be non-significant, which qualifies the descriptive dilution-effect discussion in Section 3.2. The Random Forest classification accuracy (Section 3.4) should be read cautiously, since the Classification label was itself derived from a subset of the model's input variables, so the reported accuracy reflects the labeling rule more than a validated, independent detection capability. The location-wise analysis in Section 3.6 uses randomly assigned locations with a manually specified risk weighting and should not be read as evidence about real milk quality in specific Karnataka districts. The dataset also does not contain longitudinal or microbiological measurements required to directly investigate spoilage dynamics, so all spoilage-related discussions remain derived from published literature. Finally, the reported correlations, though now statistically tested, should still be interpreted as statistical associations and not as evidence of causal relationships between the physicochemical variables.

Future work: Building on the Scope section in Chapter 3, the most valuable next steps would be validating these patterns against real, laboratory-measured milk samples; refitting the location-risk weighting against real surveillance data; and re-running the classification exercise with an independently defined quality label and proper cross-validation reporting.

Overall, this report is intended as a modest, transparently-documented contribution: a worked example of constructing and analyzing a representative milk-quality dataset, rather than a definitive empirical study of the dairy industry.

4.1 Experience and Learning Outcomes

Working on this project gave me hands-on experience with the full arc of a small applied-statistics project: constructing a dataset from literature-derived ranges, applying SMOTE to address class imbalance, running exploratory data analysis, Pearson correlation analysis, and subsequently formal hypothesis testing (ANOVA, t-tests, chi-square) to move beyond descriptive claims. Using Python for this workflow (pandas for data handling, seaborn/matplotlib for the heatmap and comparison figures, scipy for significance testing, and scikit-learn for the Random Forest exercise) let me move from raw variable ranges to a documented, analysis-ready, and statistically tested dataset relatively efficiently.

Putting this report together was also a useful lesson in the importance of documenting a dataset's provenance clearly, and in recognizing when a model's high accuracy is a product of how its labels were constructed rather than genuine predictive signal. Realizing, in revising this report, that the difference between a synthetic and a measured dataset, and between a validated result and a circular one, changes how every downstream claim should be phrased was probably the single most useful methodological lesson from this project.

Overall, this project strengthened my practical skills in exploratory data analysis, correlation-based and inferential statistical reasoning, and structuring a Python data-analysis workflow, and gave me a clearer sense of the gap between an exploratory academic exercise and a validated, industry-ready analysis.

Appendix A

Dataset

This appendix contains supporting screenshots referenced from Chapter 3: a spreadsheet view of the constructed dataset.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Location	Source	Fat	SNF	Lactose	Protein	Density	pH	AddedWat	AntibioticP	FreezingPc	Conductivi	Classification	
2	Mandya	Cow	4.2	8.78	4.53	3.29	1030.5	6.46	0.88	0	-0.488	4.08	Standard	
3	Mandya	Cow	3.94	8.8	4.59	3.49	1027.2	6.4	8.52	0	-0.544	4.01	Standard	
4	Bengaluru	Cow	4.26	8.5	4.85	3.1	1030.7	6.45	8.92	0	-0.495	4.81	Standard	
5	Shivamogg	Cow	4.61	8.77	4.82	3.4	1031.7	6.76	3.54	0	-0.526	5.23	Standard	
6	Bengaluru	Cow	3.91	8.79	4.7	3.19	1029.8	6.46	0.03	0	-0.543	4.92	Standard	
7	Shivamogg	Cow	3.91	8.49	4.72	3.14	1032.8	6.99	1.69	0	-0.54	4.67	Standard	
8	Bengaluru	Cow	4.63	9.26	4.96	3.28	1027.5	6.67	4.8	0	-0.536	4.68	Standard	
9	Davanager	Cow	4.31	8.84	4.58	3.09	1026.5	6.63	19.87	0	-0.522	4.76	Low Quality	
10	Hubballi-D	Cow	3.81	8.34	4.81	3.19	1025.4	6.47	5	0	-0.52	4.73	Standard	
11	Shivamogg	Cow	4.22	8.9	4.66	3.06	1032	6.71	1.5	0	-0.541	4.58	Standard	
12	Mangaluru	Cow	3.81	8.41	4.66	3.69	1030.3	6.51	5.03	0	-0.53	4.53	Standard	
13	Mandya	Cow	3.81	8.94	4.92	3.31	1028.9	6.62	3.88	0	-0.487	4.48	Standard	
14	Bengaluru	Cow	4.1	9.05	4.87	3.16	1029.6	6.98	7.57	0	-0.495	4.27	Standard	
15	Mandya	Cow	3.23	8.45	4.86	3.34	1026.7	6.59	0.57	0	-0.521	4.42	Standard	
16	Bengaluru	Cow	3.31	8.99	4.96	3.28	1033.9	6.77	7.16	0	-0.529	3.99	Standard	
17	Kalaburagi	Cow	3.78	8.82	4.7	3.26	1029.3	6.49	3.9	0	-0.528	4.47	Standard	
18	Kalaburagi	Cow	3.59	8.95	4.84	3.42	1029.2	6.59	16.48	0	-0.518	4.2	Low Quality	
19	Hassan	Cow	4.13	9.27	4.64	3.45	1030.5	6.87	2.09	0	-0.55	4.17	Standard	
20	Bengaluru	Cow	3.64	8.63	4.76	3.19	1030	6.51	5.01	0	-0.535	4.55	Standard	
21	Hassan	Cow	3.44	8.47	4.67	3.18	1029.4	6.87	13.44	0	-0.563	4.92	Standard	
22	Davanager	Cow	4.59	8.43	4.72	3.24	1027.4	6.71	0.54	0	-0.522	4.78	Standard	
23	Tumakuru	Cow	3.91	8.46	4.82	2.84	1029.9	6.52	13.84	0	-0.517	4.03	Standard	
24	Mysuru	Cow	4.03	8.68	4.54	3	1032.8	6.69	5.82	1	-0.54	4.2	Low Quality	
25	Mysuru	Cow	3.43	8.8	5.12	3.57	1031.7	6.75	0.35	0	-0.499	4.78	Standard	
26	Mandya	Cow	3.78	8.78	4.5	3.63	1032.2	6.69	1.79	0	-0.555	4.21	Standard	
27	Shivamogg	Cow	4.04	8.95	4.46	3.25	1028	6.36	6.16	0	-0.559	4.43	Fermented	
28	Bengaluru	Cow	3.54	8.7	4.93	3.42	1027	6.49	0.35	0	-0.526	4.67	Standard	
29	Davanager	Cow	4.15	9.14	4.86	3.36	1028.7	6.56	4.36	0	-0.509	4.21	Standard	
30	Belagavi	Cow	3.76	8.62	4.82	3.92	1029.1	6.59	2.12	0	-0.496	4.53	Standard	

Figure A.1: Spreadsheet view of the constructed dataset (see Table 2.1 for variable definitions)

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